■ Netflix Data Analysis Using Python

Internship Project Report

Data Analysis Using Python

Project	Netflix Data Analysis
Dataset Size	8,790 rows × 10 columns
Tasks Completed	6 / 6
Movies	6,126
TV Shows	2,664
Primary Tools	Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter
Author	Swayam Vekariya

This report documents the completed Netflix data-analysis internship project, covering data cleaning, exploratory analysis, visualization, trend analysis, country-wise analysis, rating and genre analysis, and business insights.

1. ■ Project Objectives

The project was developed to apply practical Python data-analysis skills to a Netflix content dataset. The main objectives were to clean the data, explore content patterns, create visualizations, identify trends, and summarize useful business-oriented insights.

Key objectives include: dataset inspection and cleaning; Movie vs TV Show analysis; country-wise analysis; release-year trend analysis; rating and genre analysis; and creation of a final business dashboard.

2. ■ Dataset Overview

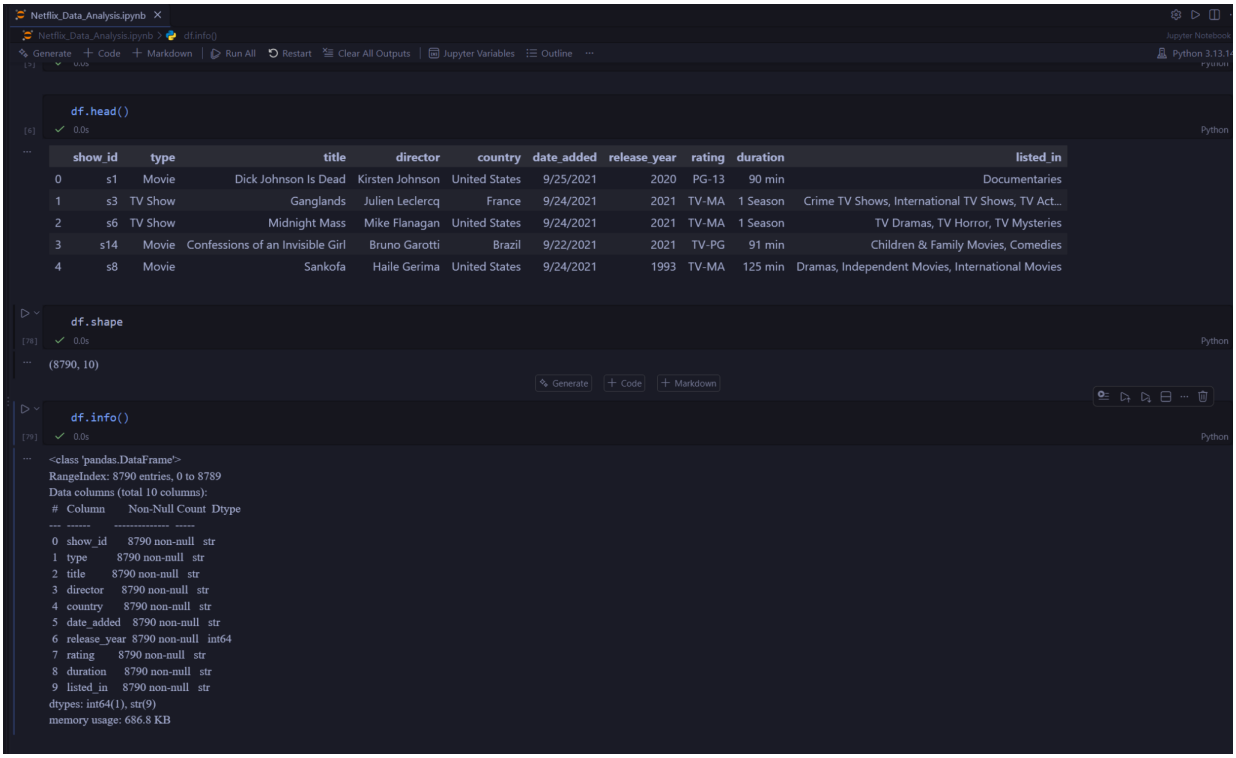
The dataset contains **8,790 records** and **10 columns**. The columns include show_id, type, title, director, country, date_added, release_year, rating, duration, and listed_in.

3. ■ Task 1 – Netflix Data Cleaning & Preparation

The dataset was imported with Pandas and inspected using head(), shape, info(), missing-value checks, and duplicate checks. The project then standardized selected text fields and exported a cleaned CSV file.

Check	Result
Rows × Columns	8,790 × 10
Total Missing Values	0
Duplicate Records	0
Cleaned Output	Cleaned_Netflix_Dataset.csv

Task 1 Evidence



```
df.head()
```

	show_id	type	title	director	country	date_added	release_year	rating	duration	listed_in
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	United States	9/25/2021	2020	PG-13	90 min	Documentaries
1	s3	TV Show	Ganglands	Julien Leclercq	France	9/24/2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...
2	s6	TV Show	Midnight Mass	Mike Flanagan	United States	9/24/2021	2021	TV-MA	1 Season	TV Dramas, TV Horror, TV Mysteries
3	s14	Movie	Confessions of an Invisible Girl	Bruno Garotti	Brazil	9/22/2021	2021	TV-PG	91 min	Children & Family Movies, Comedies
4	s8	Movie	Sankofa	Haile Gerima	United States	9/24/2021	1993	TV-MA	125 min	Dramas, Independent Movies, International Movies

```
df.shape
```

```
(8790, 10)
```

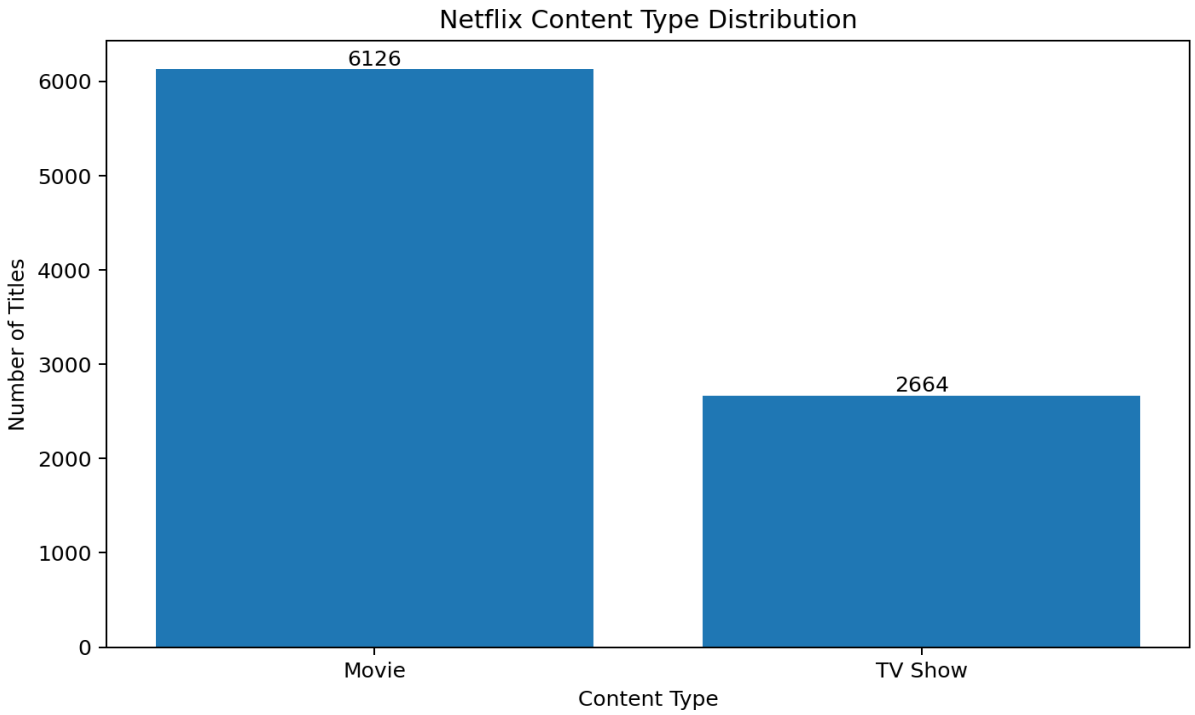
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 8790 entries, 0 to 8789
Data columns (total 10 columns):
 #  Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0  show_id       8790 non-null   str
 1  type          8790 non-null   str
 2  title         8790 non-null   str
 3  director      8790 non-null   str
 4  country       8790 non-null   str
 5  date_added    8790 non-null   str
 6  release_year  8790 non-null   int64
 7  rating        8790 non-null   str
 8  duration      8790 non-null   str
 9  listed_in     8790 non-null   str
dtypes: int64(1), str(9)
memory usage: 686.8 KB
```

4. ■ Task 2 – Content Type Analysis Dashboard

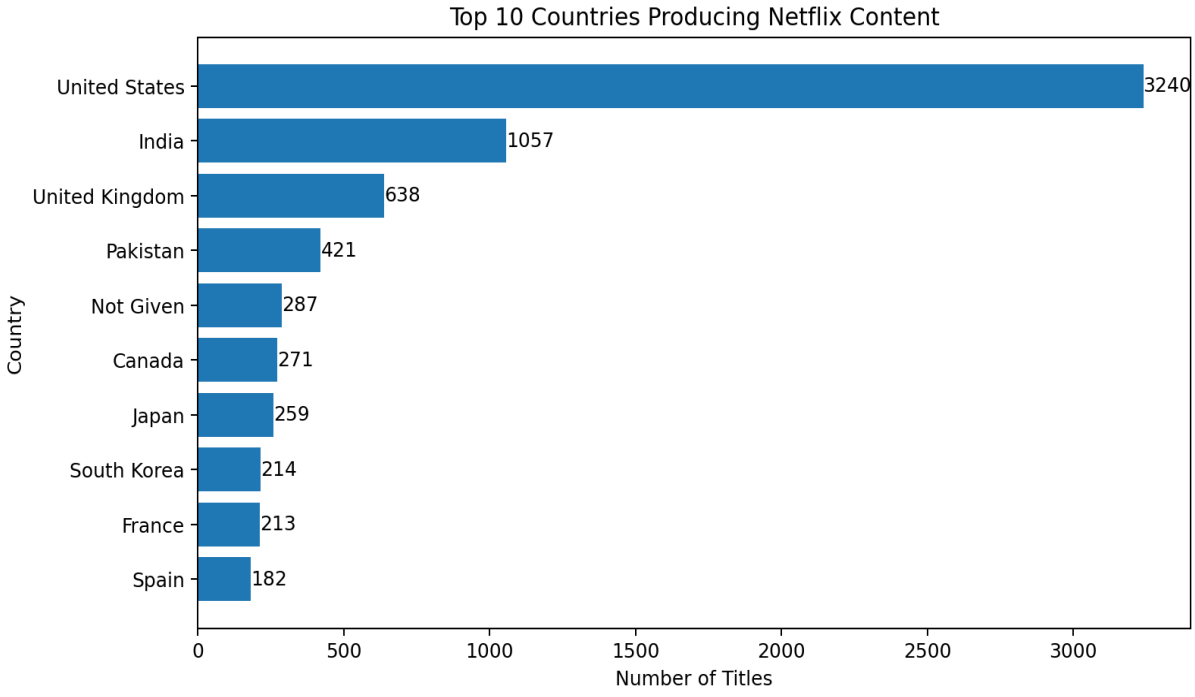
The content type analysis compared Movies and TV Shows and produced bar and pie-chart visualizations.

Movies: 6,126 **TV Shows:** 2,664



5. ■ Task 3 – Country-Wise Netflix Content Analysis

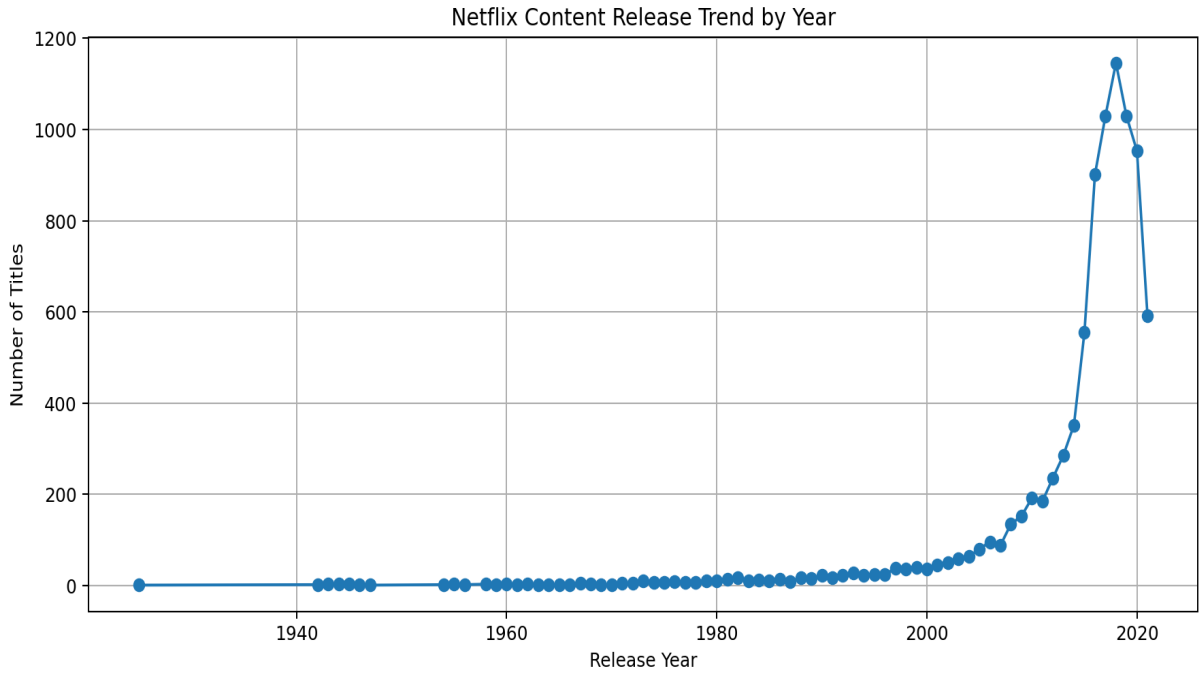
Country values were split into individual countries, cleaned, exploded into separate rows, and counted. The analysis identified the leading content-producing countries.



Top country: United States (3,240 titles).

6. ■ Task 4 – Trend Analysis by Release Year

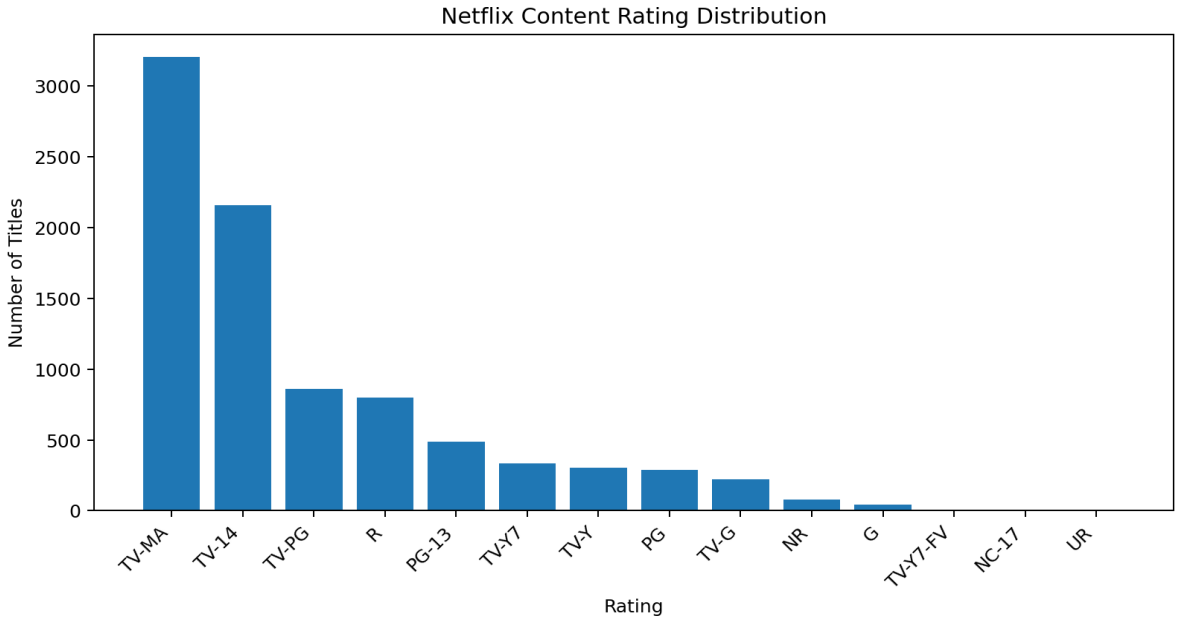
The project counted Netflix titles by release year and visualized the resulting trend to identify periods with higher content volume.

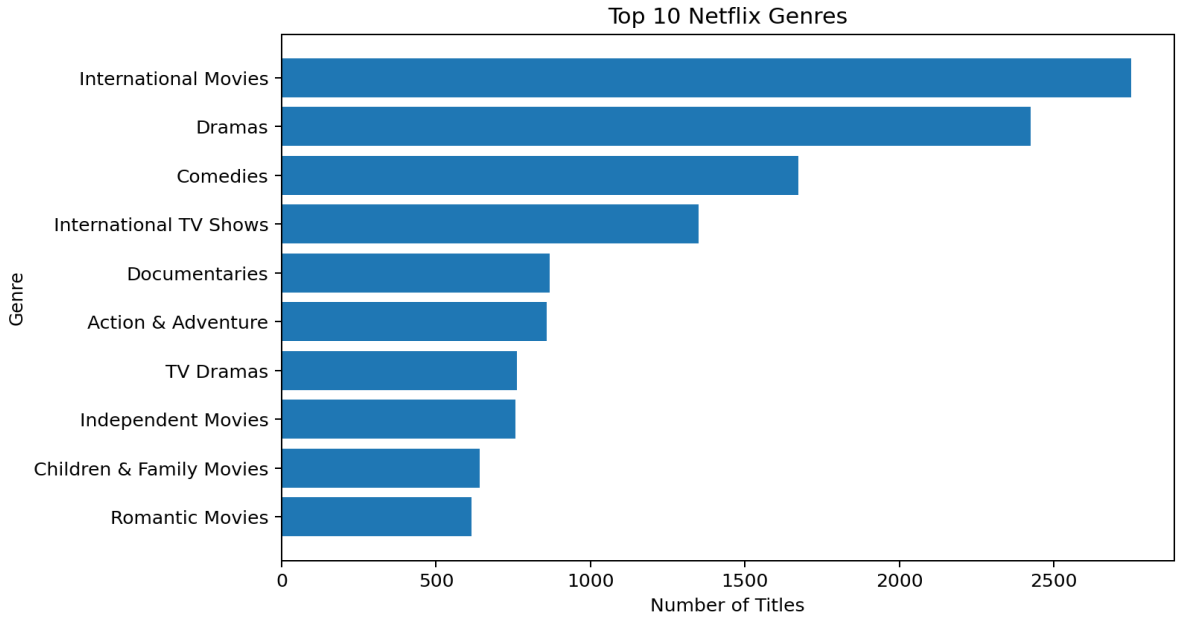


Peak release year in the dataset: 2018 with 1,146 titles.

7. ■ Task 5 – Content Rating & Genre Analysis

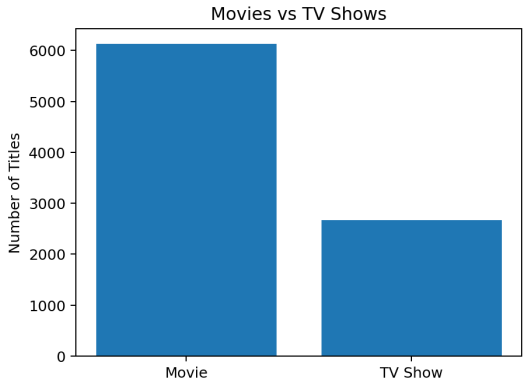
The analysis examined content ratings, extracted individual genres from the listed_in field, ranked the most common genres, and compared ratings across Movies and TV Shows.

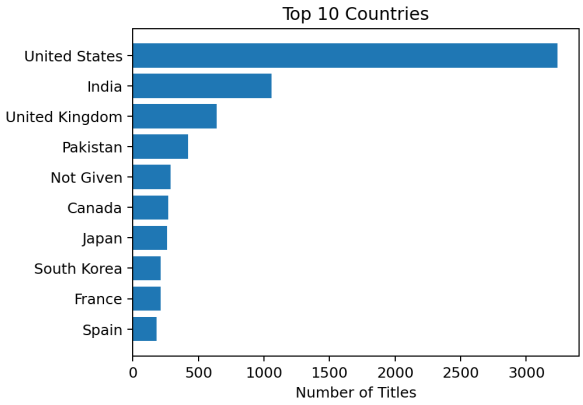


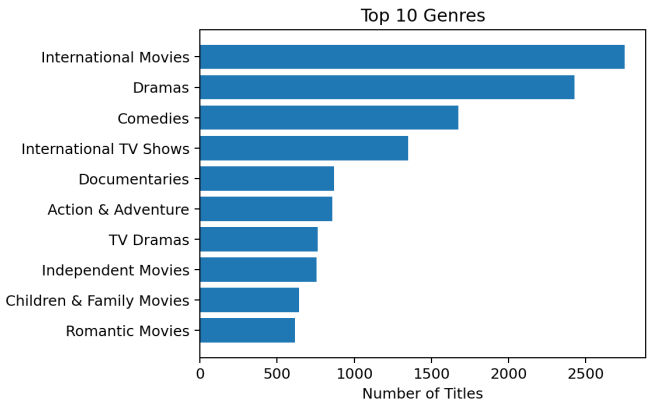

Top 10 Netflix Genres
International Movies
Dramas
Comedies
International TV Shows
Documentaries
Action & Adventure
TV Dramas
Independent Movies
Children & Family Movies
Romantic Movies
Genre
0
500
1000
1500
2000
2500
Number of Titles

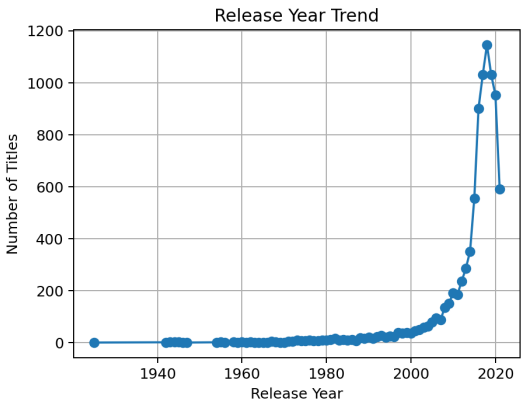
8. ■ Task 6 – Netflix Business Insights Report

A final dashboard combined the major analysis areas into one visual summary. The dashboard includes content type distribution, top countries, top genres, and release-year trends.









Metric	Result
Total Titles	8,790
Movies	6,126
TV Shows	2,664
Most Common Rating	TV-MA
Most Common Genre	International Movies
Top Producing Country	United States
Peak Release Year	2018

9. ■ Conclusion & Learning Outcome

The completed project demonstrates an end-to-end Python data-analysis workflow: loading a real-world dataset, checking data quality, preparing fields for analysis, performing exploratory analysis, creating visualizations, identifying trends, and presenting business-oriented findings.

The project also demonstrates practical use of Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook, and structured reporting for an internship submission.

■■■ Author: **Swayam Vekariya**
■ BCA Graduate | ■ Aspiring Data Analyst / Python Developer

■ Thank You